Permission Examples

DEVOPS-002 · Task 4 - Linux File Permissions · Sprint 9 (29/6 - 12/7)

Run each **chmod** command below in your terminal, then run **ls -l file1.txt** right after it. The expected output under each command shows what your result should match.

Understanding Read, Write, Execute

Every file or directory in Linux carries three permissions, applied separately to the **owner**, the **group**, and **others**:

Permission	Symbol	Meaning
Read	r	View a file's contents, or list a directory's contents
Write	w	Modify a file, or add/remove files inside a directory
Execute	x	Run a file as a program, or enter (cd into) a directory

Each numeric digit in a **chmod** command is the sum of the rights it grants: **r = 4, w = 2, x = 1**. The three digits, in order, always represent **owner, group, others**.

Quick Reference

Octal	Symbolic	Owner	Group	Others	Typical use
777	rwxrwxrwx	rwx	rwx	rwx	Full access for everyone (rarely appropriate)
755	rwxr-xr-x	rwx	r-x	r-x	Scripts / programs / directories
644	rw-r--r--	rw-	r--	r--	Regular files and documents
600	rw-------	rw-	---	---	Private files, only owner access
400	r--------	r--	---	---	Read-only sensitive files (e.g. SSH keys)

1. chmod 777 file1.txt

COMMAND

```
chmod 777 file1.txt
```

EXPECTED OUTPUT (ls -l file1.txt)

```
-rwxrwxrwx 1 trainee1 developers 0 Aug 7 09:15 file1.txt
```

Explanation: Grants full read, write, and execute access to everyone - owner, group, and others. This is the most permissive setting and is rarely appropriate outside temporary testing, since anyone on the system can modify or run the file.

2. chmod 755 file1.txt

COMMAND

```
chmod 755 file1.txt
```

EXPECTED OUTPUT (ls -l file1.txt)

```
-rwxr-xr-x 1 trainee1 developers 0 Aug 7 09:16 file1.txt
```

Explanation: The owner gets full access (read, write, execute); the group and others can read and execute but not modify. Common for scripts, programs, and directories that need to be run or browsed by others but only edited by the owner.

3. chmod 644 file1.txt

COMMAND

```
chmod 644 file1.txt
```

EXPECTED OUTPUT (ls -l file1.txt)

```
-rw-r--r-- 1 trainee1 developers 0 Aug 7 09:17 file1.txt
```

Explanation: The owner can read and write; the group and others can only read. This is the standard, safe default for regular documents and configuration files that don't need to be executed.

4. chmod 600 file1.txt

COMMAND

```
chmod 600 file1.txt
```

EXPECTED OUTPUT (ls -l file1.txt)

```
-rw------- 1 trainee1 developers 0 Aug 7 09:18 file1.txt
```

Explanation: Only the owner can read or write the file - group and others have no access at all. Used for private or sensitive files, such as personal notes or credentials.

5. chmod 400 file1.txt

COMMAND

```
chmod 400 file1.txt
```

EXPECTED OUTPUT (ls -l file1.txt)

```
-r-------- 1 trainee1 developers 0 Aug 7 09:19 file1.txt
```

Explanation: Only the owner can read the file, and even the owner cannot write to it without changing permissions again first. Typically used for highly sensitive read-only files, like SSH private keys.